

Movie Recommendation System Report

Introduction

The Movie Recommendation System is a web-based application that suggests movies to users based on genre similarity. The system uses Machine Learning techniques to compare movies and provide top recommendations. The project is developed using Python and Streamlit for an interactive user interface.

Abstract

This project implements a content-based movie recommendation system using movie genres as the primary feature. The application processes movie data, converts genres into numerical vectors using CountVectorizer, and calculates similarity between movies using cosine similarity. Users can select a movie and instantly receive five recommended movies with similar genres.

Tools Used

- **Python** – Core programming language
- **Pandas** – Data handling and processing
- **Scikit-learn** – Machine learning functions
- **Streamlit** – Web application framework
- **CountVectorizer** – Converts text data into vectors
- **Cosine Similarity** – Measures similarity between movies

Steps Involved in Building the Project

1. Load the movie dataset using Pandas.
2. Clean and preprocess the genres column.
3. Convert genres into vectors using CountVectorizer.
4. Calculate similarity scores using cosine similarity.
5. Create a recommendation function to suggest similar movies.
6. Build the user interface using Streamlit.
7. Display the top 5 recommended movies to the user.

Conclusion

The Movie Recommendation System successfully recommends movies based on genre similarity. The project demonstrates the practical implementation of Machine Learning concepts such as text vectorization and similarity measurement. It also provides a simple and user-friendly interface for users to explore movie

recommendations.